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RESEARCH

Detection of hateful twitter users with graph convolutional network model

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Abstract
Today, hate speech is widespread and persistent in various forms on social networking platforms, targeting different minority groups. These attacks can be carried out using various factors such as racial, religious, gender, and physical disability, etc. Considering the number of people and their interactions, social networks are the most important channels through which these discourses spread. The social network structure is considered a set of nodes and edges and is very suitable for the graph structure. The multidimensional structure of social networks carries social network data from Euclidean space to non-Euclidean space. In non-Euclidean space, the graph structure is used to represent data effectively. In this respect, solving the hate speech problem with graph-based methods in a complex dimensional space can produce more impressive results. In this study, a powerful method based on the Graph Convolutional Network (GCN) model, which is rarely used in this field, was proposed for the detection of hateful Twitter users in social networks. Well-known machine learning methods were used to measure the performance of this method. According to the results obtained, the proposed GCN model gave the most successful result.

Keywords Hate speech detection · Graph convolutional network · Deep learning · Machine learning

Introduction

Due to the circumstances of the age we live in, social media has a significant role in our lives. On social media platforms that have a huge impact on society, users can freely share their ideas, feelings, and thoughts. This freedom also brought sharing of hateful and disturbing content (Bölücü and Canbay 2021). Although there is no general definition of Hate Speech (HS), it is a social media problem that can be considered a criminal attempt seen in all scenarios where social media users can intentionally or unintentionally make messages or comments that may insult an individual or a segment of society (MacAvaney et al. 2019). It can also be defined as a language that encourages, incites, or increases violence against certain groups based on certain characteristics. HS should not be confused with freedom of expression. Because although social media is interpreted as free thought sharing when the purpose of the use is considered, freedom of expression and press must be absolute. Posts encouraging violence in social media environments are considered violent crimes today (Gitari et al. 2015). Detection of such shares is very important for public authorities and social media networks.

The history of social networks is as old as the human species. The transition to an agricultural society and growing urban civilizations made the networks between human communities larger and more complex (Knoke and Yang 2019). While social networks were mainly the field of social sciences at first, it has now become an interdisciplinary research field with the contributions of Sociology, Social Psychology, Anthropology, Physics, Mathematics, Computer Science, and other disciplines. In a study conducted in 1978 to understand the social network structure among scientists

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working in various departments at a famous university in the USA, the social network formed among the astronomy department itself is shown in Fig. 1. In the seventies, technological possibilities were limiting the analysis of large amounts of data and therefore large-scale networks. Later, with the internet revolution, technology and communication opportunities increased enormously. This revolution enabled the establishment of large-scale social networks by interacting socially with millions online. From this point on, online network structures that we call Online Social Networks (OSNs) were formed.

Social networking environments provide new possibilities for presenting personal expressions, creating communities with common interests, collaboration, and sharing (Murray 2008). OSN sites are web-based services that allow users to create public or semi-public profiles, add other people they are connected to their friend lists, and visit and review the profiles of the people on their lists. Social networks attract millions of users thanks to their features that allow sharing of a wide variety of content and profile information such as photos, videos, and texts. Therefore, the use of social networks among internet users is spreading daily. For example, the number of users of Facebook, the popular OSN platform, approached three billion, and YouTube reached two and a half billion users as of the beginning of 2022 (Most popular social networks worldwide as of January 2022, ranked by the number of monthly active users 2022).

Twitter is today's most popular social media platform that prioritizes sharing meaningful content where people share their feelings and opinions about certain topics (Nielsen 2011). Data on Twitter is huge and unstructured. For these reasons, it is difficult for people to analyze Twitter data and obtain subjective information. Recently, sentiment analysis was applied in various fields such as commercial, political, and social media. Sentiment analysis methods, as in many social media platforms, are the most important tools that enable us to obtain the feelings and opinions of individuals from Twitter data. Through sentiment analysis, it can help

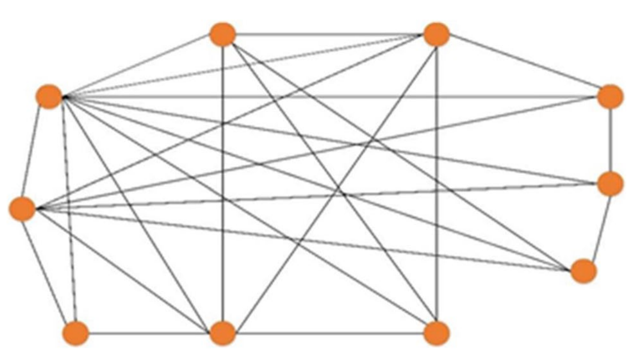


Fig. 1 The social network structure of the statistics department (Friedkin 1978)

organizations such as companies and even governments to gain insight into people's views on their decisions. Sentiment analysis and Idea Mining led to an increase in research and techniques especially in the field of Natural Language Processing (NLP) in the last decade. Although it mostly investigates the feelings and thoughts of users about a certain product or an event, research was carried out recently to detect many situations that may negatively affect social and ethical issues such as fake news, extremism, violence, and social media environments. HS, also called toxic hate speech, is a phenomenon that can easily spread in social media environments. In recent years, social media tools became frequently used tools for the dissemination of both authorized information and misinformation. It is a prime example of how online media's excellent dissemination trend turns into both opportunities and challenges. For this reason, HS detection in social media has recently become the focus of attention of many researchers doing research in the field of NLP. Developing automatic tools can be one of the most critical measures to prevent dangerous tendencies such as the escalation of violence and hatred.

Social media platforms and media tools should provide proven, unbiased information that is constructive and positive instead of providing false information that will divide society and create negative emotions such as fear, anxiety, and stress. In this perspective, the purpose of HS detection is to distinguish hateful users from non-hateful users. There were many studies examining the analysis of such shared information and HS detection is a widely researched task with many challenges (Poletto et al. 2021). The majority of studies in the literature analyzed social network posts containing hate speech using various methods to determine whether the post contains hate speech or not. In this study, in addition to analyzing tweeter posts, hateful users were detected by using users' features such as tweeting frequency, number of followers, number of favorites, and retweets. Therefore, a graph-based GCN embedding approach was proposed to detect Hateful users on Twitter. The GCN method allows us to analyze Twitter data more comprehensively than traditional machine learning models. Here, Decision Tree (DT), Linear Regression (LR), Naïve Bayes (NB), Random Forest (RF), Support Vector Machine (SVM), Multilayer Perceptron (MLP), and Bidirectional Long-Short Term Memory (Bi-LSTM) methods were used to measure the performance of the proposed GCN method. GCNs are neural networks that can be applied directly to graphs and provide an easy way to perform node-level, edge-level, and graph-level prediction tasks. GCNs can do what CNNs cannot. Experimental results prove the success of the proposed model. The results obtained from this study will make valuable contributions to the detection of HS discourses that trigger feelings and thoughts of hatred, violence, fear, and aggression in the public in social media environments. It can also help government agencies and social media platform providers plan early to take action against toxic HS.

Motivation and contribution

With the developing technology, the views expressed on Twitter, one of today's most important social media platforms, often cause controversy. Different opinions and controversial content directly affect interactions between communities. Recently, controversial content was replaced by offensive content and hate speech. Automatically detecting users posting such content that can cause social unrest is challenging. Most of the existing approaches in the literature focus on the content of hateful messages (Mulki et al. 2019; Albadi et al. 2018; Roy et al. 2020). To overcome this limitation, we propose a novel GCN-based approach for hateful user detection by leveraging the power of Twitter data to use both the message content and the structured network features of Twitter users.

The main contributions of the present study are summarized as follows:

- In the proposed model, the ability of GCNs to encode relational data is exploited to create a network embedding vector model that combines tweet content and Twitter user information. Since the generated network has both semantic and structural properties, it contains more comprehensive information for HS detection compared to classical models.
- With the GCN network model, each node collects the feature information of its neighboring nodes in the convolution layer. Then, a combined feature set is obtained. New results are obtained by applying non-linear transformations to the obtained features. Also, a three-stage GCN was used to extract more meaningful features in the proposed model. In the proposed GCN network model, the output of the convolutional layer feeds the next convolutional layer as input. This way, similar features between nodes are captured while transferring information between layers. The obtained information provides a significant advantage when identifying Hateful users.
- Experiments show the positive effect of combining textual features and structural information in a graphic, in terms of performance and accuracy.

Related work

Considering the effect of online social media environments on the spread of HS, the number of research and techniques produced in the literature is increasing day by day. There are many studies in this field, many datasets used and different techniques proposed. Most of the work done to detect HSs is based on Text Mining techniques. Poletto et al. (2021) presented a detailed analysis of the methods proposed in the literature for the detection of HS. The authors of the study; analyzed suggested models, dictionaries, and resources used in different categories. Many studies analyze methods for detecting HS like this (Al-Hassan and Al-Dossari 2019; Naseem et al. 2021; Cao et al. 2020).

The studies proposed in the literature generally range from traditional machine learning models based on dictionary-based approaches to new methods based on deep learning such as convolutional neural networks (CNN) and recurrent neural networks (RNN) using attention mechanisms (Badjatiya et al. 2017; Gröndahl et al. 2018). There are also various HS detection studies that were developed using machine learning algorithms for different languages in the literature. Rangel et al. (2021) tried to identify possible spreaders of hate in online environments by analyzing the posts of social media phenomena. To this end, they presented hate speech word corpuscles based on English and Spanish languages. In their study, Abro et al. (2020) proposed machine-learning approaches using datasets with three different class labels for HS detection. The authors achieved the best performance in the experimental results with the bi-gram and support vector machine (SVM) classifier. They also emphasized that they will use different feature engineering techniques and artificial intelligence algorithms in their future studies to achieve higher performance. Gitari et al. (2015) proposed a new dictionary-based approach for HS detection in their study. It is a new dictionary that focuses on the extraction of subjective features that the authors created with a rule-based approach in their work. Experimental results revealed the addition of semantic and topic-based features to the proposed dictionary-based approach. Nobata et al. (2016) proposed a machine learning-based approach for detecting abusive language on online platforms, which analyzes different levels of abusive language over time. In addition, the authors created a community of user comments for abusive language. A new supervised attention approach based on keyword extraction and word graph based on BERT's attention mechanism was proposed for HS detection (Sarracén and Rosso 2021). When extracting keywords, the authors proposed a model that uses the differences between hateful and non-hateful texts and the harmonic mean of the words' relative frequencies. Roy et al. (2020) proposed a deep learning-based approach based on the deep convolutional neural network model (DCNN) for HS detection using a shared Twitter dataset in English. In the proposed model, the GLoVe word embedding approach is used for word representation. MacAvaney et al. (2019) proposed a new approach based on SVM. In their analysis, the authors argued that SVM would be simpler and easier to interpret than neural network methods involving attention mechanisms. Garain and Basu (2019) used the RNN-based

Bi-LSTM deep learning approach in the model they developed for HS detection using the Twitter dataset on women and refugees on Twitter. The main disadvantages of the proposed model are that the authors used a single classifier in the experimental results and they used any data other than the dataset they used. Zimmerman et al. (2018) proposed a new approach in their study, in which they combined ten CNNs with different initial weights into a community to maximize the power of CNNs to improve HS detection.

In Table 1, the results of the proposed GCN-based model were compared in terms of accuracy metric with the studies conducted in the literature to detect hate speech on Twitter. It can be observed that the proposed approach displayed a higher classification performance compared to other proposed approaches in the current literature.

Graph Neural Networks (GNNs) are deep learning-based approaches used to manipulate data that can be represented as graphs. The most basic part of GNNs is a graph. GNNs became a powerful tool for many important applications as graphs become more common and rich with information and neural networks become more popular and capable. In various studies, researchers used the GNN graph structure, which is effectively used in the solution of non-regular, multidimensional, and complex problems in non-Euclidean data space. Zhang et al. (2019) categorized the first studies with GNNs in the literature as (Recurrent Graph Neural Networks) RecGNNs. However, the high computational cost of these models emerged as an important issue. To solve this problem, the researchers used a new approach called convolutional GNNs (GCN), which can be applied to graphs. While developing this method, the success of CNNs in image processing has been a source of inspiration. GCNs are divided into spectral-based and spatial-based. In the spectral-based approach, the models are based on spectral graph theory and the signals in a graph are filtered using the intrinsic composition of the Laplacian graphs. In the spatial-based approach, on the other hand, convolution operations are performed directly on graphs, and graph convolution is represented as a combination of feature information from node neighbors (Zhang et al. 2019). Later, Wu et al. (2020) proposed a new taxonomy. Accordingly, GNN was divided into four groups: recurrent GNNs (RecGNN), convolutional GNNs (ConvGNNs), graph autoencoders (GAEs), and spatial–temporal GNNs (STGNNs).

Today, it is a method used especially for the analysis of social networks. In particular, Graph Autoencoders (GAE) is unsupervised learning frameworks that encode nodes and graph structures in social networks into a hidden vector space. To this end, they reconstruct the original graphic inputs into word insertion vectors. In doing this, a graphic taken as input via an encoder is first converted to a low-dimensional vector. Then, the vector generated in the previous step is converted to the original input via a decoder. In this way, the information loss between the input graph and the output graph is minimized (Wu et al. 2020). Peng et al. (2018) proposed a graphics-based deep learning approach to convert the text they process as input into graphics. Then, they made text classification using GCNs. Mishra et al. (2019) proposed a GCNs model that captures the linguistic behavior of social media users for abusive language detection on online social media platforms. This proposed model is one of the first approaches in the literature to use GCNs for the detection of HS.

Table 1 Comparison of the proposed GCN-based model with other models in the literature

Model	Accuracy	Reported by
The Proposed GCN based Model	0.930	Ours
NB	0.903	Mulki et al. (2019)
SVM	0.832	
GRU-based RNN	0.790	Albadi et al. (2018)
Sentiment-based, Semantic Unigram	0.754	Watanabe et al. (2018)
BiLSTM	0.752	Naseem et al. (2021)
Logistic Regression	0.750	Dukic and Krzic (2021)
SVM	0.790	Abro et al. (2020)
AdaBoost	0.780	
Rule based	0.734	Gitari et al. (2015)
Voted Ensemble Classifier	0.890	Burnap and Williams (2015)
2CNN	0.920	Roy et al. (2020)
Multi-view SVM	0.803	MacAvaney et al. (2019)
BiLSTM	0.703	Garain and Basu (2019)
GraphSage	0.90	Ma et al. (2019)
GCN with additional conv layer	0.814	Bölücü and Canbay (2021)

Base models

In this study, the proposed GCN model and effective machine learning methods were used for the detection of hateful users. In this section, these methods were briefly introduced.

- *DT*: This model is an example of a decision structure with a tree shape, and its classes are determined via induction from sample data that is known (Song and Ying 2015). DTs are the most widely used method among classification and regression models. The reason for this is that decision tree methods are easy to interpret and can be easily integrated with other systems. Therefore, it is to put forward understandable rules and be reliable. DTs are generated from general to specific and downward-trained data. The tree starts with a root node containing all the data in the sample (Safavian and Landgrebe 1991). The internal nodes of the DT represent the terms and the leaf nodes represent the classes.
- *LR*: It is a method based on statistics, which is used to reveal the cause-and-effect relationship between a dependent variable and one or more independent variables. The regression model associates the dependent variable with the independent variable or variables through a function (Luu et al. 2021). A single argument can be used in the LR method, or multiple arguments can be used. When more than one independent variable is used, it is called multiple linear regression.
- *NB*: It is a simple probability-based classification method based on Bayes' theorem (Rish 2001). The NB method is a simple yet powerful algorithm for predictive modeling. Among the classification methods, features such as multi-class support, categorical prediction support, prediction speed, memory usage, and interoperability are superior to other methods. It has successful applications in areas such as software defect prediction, health care, cybersecurity, and education (Wickramasinghe and Kalutarage 2021), also it has one of the most widely used classifications and prediction algorithms, especially in signal and image processing fields.
- *RF*: This effective algorithm is one of the ensemble learning methods. Ensemble learning methods aim to improve results by combining different models. The RF algorithm is obtained from a collection of multiple decision trees (Breiman 2001). The RF algorithm offers a fast, flexible and powerful structure for analyzing high-dimensional data (Antoniadis et al. 2021). It's an unusually capable algorithm that can handle thousands of variables without deleting them or degrading their accuracy. RF was successfully applied in many fields such as chemoinformatics, ecology, 3D object recognition, bioinformatics, spam detection, credit card fraud detection, text classification, and event prediction (Biau and Scornet 2016).
- *SVM*: This method is one of the widely used supervised machine learning techniques for classification and regression and was developed by Vapnik (Cortes and Vapnik 1995). SVM is a learning method developed in the field of statistical learning theory. SVM first transfers the data to a higher dimension where it can be separated linearly. The main purpose of SVM is to find a function in a multidimensional space that can separate the training data with known class labels. SVM was applied in various fields such as image classification, speech detection, text classification, cyberbullying detection, and face detection (Pradhan 2012).
- *MLP*: It is a feed-forward neural network with one or more layers between the input and output layers. In this model, the input layer is the layer where the incoming information is received and directed to the hidden layer for the purpose of learning. The hidden layer where the learning process is performed can be one or more, and the layer where the information output is provided is called the output layer (Okwu and Tartibu 2021). Each neuron in the layers is connected to each neuron in the adjacent layers (Area and Mesra 2012). Although the number of hidden layers and the number of neurons in the hidden layer is not certain, they are two important factors that affect the quality of education.
- *Bi-LSTM*: This model is very successful in classification and predicting tasks. Bi-LSTM may learn long-term dependencies using the past and future data present in the time series. To understand past and future information, the Bi-LSTM architecture has a forward and backward layer made up of LSTM units. In doing so, it does not take into account unnecessary contextual information (Zhang et al. 2020; Jang et al. 2020).
- *GCN*: This is a multilayer neural network model running on a graph. GCN is a strong version of the GNN and was produced from Graph Spectral Theory (Chung 1997). GCN works on vectors and matrices like classical neural network models. A vector of numeric values reflecting pertinent information about nodes and their connections is produced by GCN from structured graph data. The term "graph embedding" refers to this vector representation. Machine learning frequently uses this technique to turn complex information into a structure that can be differentiated and learned. Building on the CNN model, GCN aims to take the concept of convolution beyond the simple two dimensions. Convolution takes a small subsegment of the image and applies a convolution function

to it, producing a new part. Here, the central node collects information from its neighbors and itself to generate a new value. GCN uses the properties of neighboring nodes to create an embedded node structure (Kipf and Welling 2017). GCN maps nodes into a d-dimensional embedding space so that similar nodes in the graph are placed close together. It is intended to map the nodes so that the docking area's similarity approximates the network's similarity.

Methodology and materials

Description of the model

With the increased use of the Internet and the spread of social media platforms, the Web content produced is also increasing rapidly. Social media platforms generally allow users who are connected or like-minded and have common interests to interact. However, shares that may be perceived as usual or entertaining by users may be disturbing for another user. In addition, social media platforms provide users with the opportunity to share all kinds of thoughts. Accordingly, expressions containing hate speech are also increasing.

Traditional machine learning and deep learning models need to be in a structure to model the relationships and interactions between users on social media platforms. However, modeling user interactions with a graph structure also reveal hidden connections. For this reason, the use of graph-based models comes to the fore in studies using social media data. In this study, a GCN-based detection system was proposed to detect hateful users. With the developed graph structure, it is aimed to model user interactions and connections more accurately. The proposed model was extensively compared with DT, LR, NB, RF, SVM, MLP, and Bi-LSTM using accuracy, precision, recall, and F1-score.

Dataset

In this study, a dataset created by researchers at Universidade Federal de Minas Gerais in Brazil and made publicly available on Kaggle was used. The dataset used consists of 100,368 Twitter users and their activities.

For each user, there are events such as tweeting frequency, number of followers, number of favourites, and number of hashtags. In addition, dictionary analysis derived using the last 200 tweets of each user enabled the acquisition of language content-related features. Using the Empath tool in the dataset, it was used to analyse each user's dictionary for categories such as love, violence, community, warmth, ridicule, independence, envy, and politics, and assign numerical values to indicate the user's agreement with each category.

The dataset contains 204 attributes to characterize each user as hateful or normal. There are important relationships between Twitter users in the dataset. If a user retweeted another user, those users are considered linked. About 5000 of the users in the dataset are labeled as hateful or normal. The users_anon_neighborhood.csv file included in the dataset contains the number of people tweeted for 1 neighbourhood as well as various attributes for each user. The first 5 lines and the first 10 attributes of this file are shown in Fig. 2.

The distribution of users in the dataset is 95415 other, 4427 normal and 544 hateful. Of the 1039 node features in the dataset, only 206 attributes based on the user's attributes and the tweet dictionary are used. Figure 3 presents the heatmap showing the relationships of the first 10 features in the dataset. Since there are 206 features in the dataset, the heatmap of a sample consisting of 10 features is presented in Fig. 3.

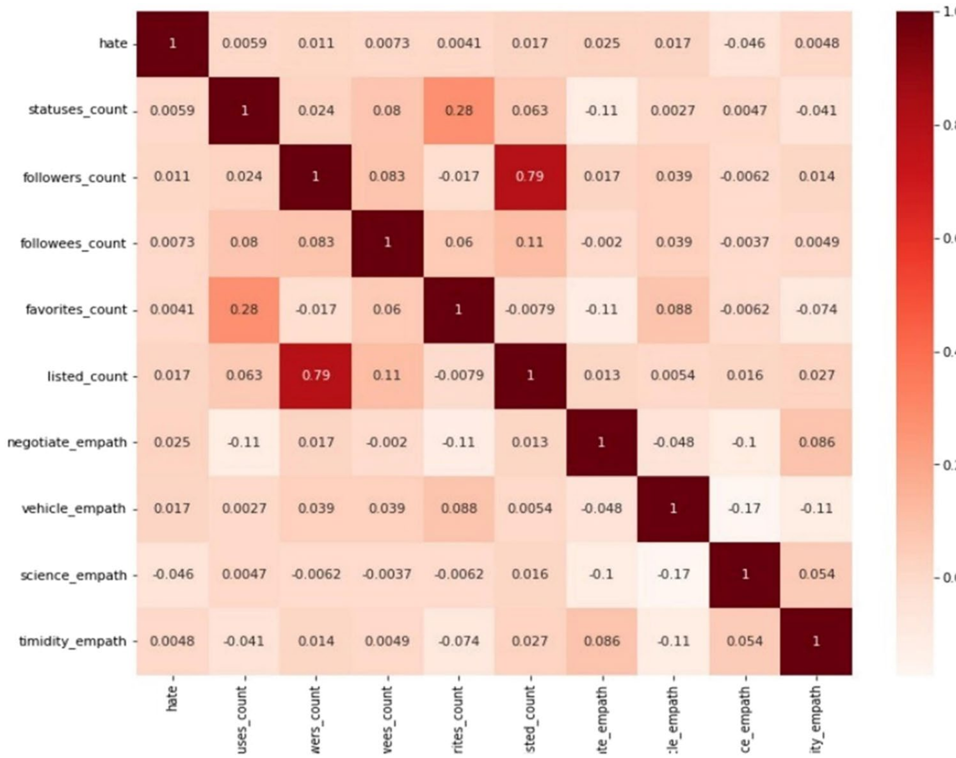
The features used are normalized. The dataset is divided into 80% for the training set and 20% for the test set. The training data is divided into 90% for training and 10% for validation. After obtaining the train and test sets, the shape of the train set (1211, 204) and the shape of the test set (303, 204) were obtained. The adjacency matrix showing the relationships between the users in the dataset is shown in Fig. 4.

A sample of the adjacency matrix is shown as so many users follow each user. The adjacency matrix for the first 5 users in the dataset is presented in Fig. 4.

	user_id	hate	hate_neigh	normal_neigh	statuses_count	followers_count	followees_count	favorites_count	listed_count	betweenness
0	0	normal	True	True	101767	3504	3673	81635	53	100467.895084
1	1	other	False	False	2352	19609	309	61	197	0.000000
2	2	other	False	False	1044	2371	2246	561	16	4897.117853
3	3	other	False	False	167172	3004	298	3242	53	9.864754
4	4	other	False	False	1998	17643	19355	485	239	0.000000

Fig. 2 The first 5 lines and the first 10 attributes of users_anon_neighborhood.csv file

Fig. 3 Heatmap of the first ten features in the dataset



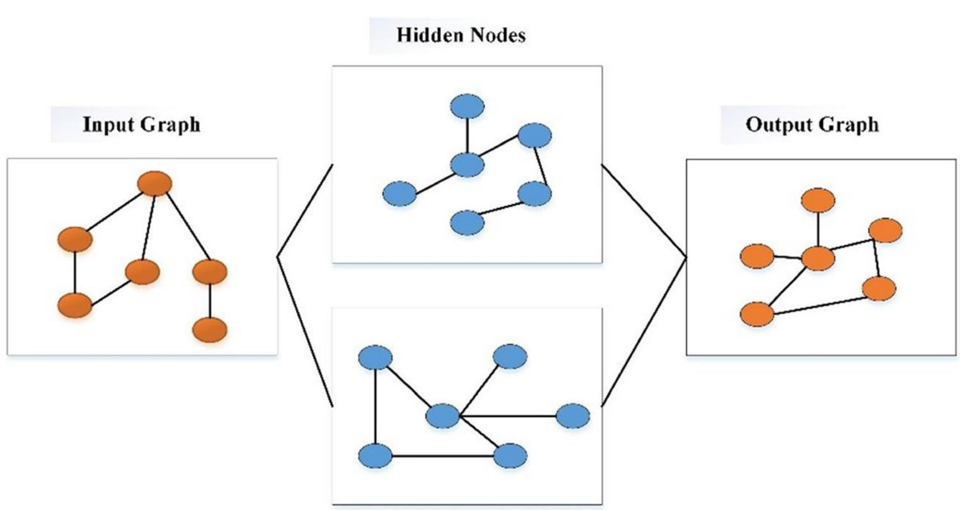
Proposed GCN based model

A sort of machine learning algorithm known as the GNN model is capable of extracting significant data from graphs and producing helpful predictions. GNNs emerged as a potent tool for numerous critical applications as graphs got more prevalent and information-rich and as neural networks gained popularity and capability. It is a widely used model today, especially for the analysis of social networks. In Fig. 5 basic structure of GNN is shown. In the GNN

Fig. 4 The adjacency matrix

	10999	…	253	2813	88427	26889	31134	…	29251	46488	…	41027	93947	…	86339
10999	1	1	1												
2813				1	1	1									
31134							1	1	1						
56761										1	1	1			
93947													1	1	1

Fig. 5 The basic structure of the GNN model



structure, the input feeds the hidden nodes of the graph and the data gains a graph-structural representation. Then an output graph is produced from this structure.

GCN is a powerful version of GNN. Unlike CNN, the GCN method deals with irregular graphs, and the data in these graphs depend on graph Laplacian matrices (Guo et al. 2021). In these, it performs the filtering operation in the frequency domain. Figure 6 shows a basic diagram of the GCN structure. Here, each convolutional layer contains a hidden representation of each node by collecting the feature information of the neighboring nodes of each node. A non-linear transformation is applied to these aggregated features and a result is obtained. After further processing of multiple layers, nodes receive information from subsequent nodes, revealing a hidden display.

The basic theoretical representation of the GCN structure is given below. Here, G represents a graph, and $G=(V, \mathcal{E}, A)$. V, $\mathcal{E}$ denotes the set of nodes and edges where $|V|=$ n and $|\mathcal{E}|=m$. A indicates the neighborhood matrix. If there is an edge between node i and node j, A(i,j) indicates the weight of the edge. Otherwise A(i,j) = 1. For unweighted graphs, A(i,j) = 1 is defined. Diagonal matrix D specifies a degree matrix where $D(i,i)=\sum_{j=1}^{n} A(i,j)$. L is the Laplacian matrix of A and is calculated as $L=D\text{-}A$. The symmetrically normalized Laplacian matrix is expressed as $\widetilde{L}=I-D^{-\frac{1}{2}}AD^{-\frac{1}{2}}$. Here I represent the identity matrix. The graph signal at a node is represented as the $x \in R^{n}$ vector. The signal value at node i is expressed by $x(i)$. The node attributes are considered a graph signal, and the matrix of an attribute graph is represented by $X \in R^{nxd}$. Columns of X are d signals of a graph (Zhang et al. 2019).

Transform of graph Fourier

Considering the Fourier transform of the 1-D f signal, the Laplacian Matrix L is the Laplace operator in a graph. From this perspective, an eigenvector of L is analogous with exponential complexity at a certain frequency. The normalized matrix $\widetilde{L}$ refers to the graph Laplace operator. The eigenvalue decomposition of $\widetilde{L}$ is shown as $\widetilde{L}=U \wedge U^{T}$. The lth column of U is the Eigenvector u_l and $\wedge(l,l)$ is the appropriate Eigen value λ_i. From here, the Fourier transform of an x-graph signal is calculated as follows (Zhang et al. 2019):

$$\hat{x}(\lambda_l)=\sum_{i=1}^{n} x(i)u_l^{*}(i) \quad (1)$$

Filtering of graph

Graph filtering is performed on graph signals. A signal can be filtered at the node or spectral field of a graph.

a) Frequency filtering: The structure of graphs is generally irregular, but graph convolution in the vertex areas of a graph is easier than classical signal convolution in the time domain. Spectral graph convolution is defined as (Zhang et al. 2019):

$$(x * Gy)(i)=\sum_{l=1}^{n} \hat{x}(\lambda_l)\hat{y}(\lambda_l)u_l(i) \quad (2)$$

$\hat{x}(\lambda_l)\hat{y}(\lambda_l)$ specifies filtering in the spectral field. The filtering of an x signal in the G graph with the y filter is shown by Eq. 2.

b) Vertex Filtering: The vertex filtering of the x signal at an i node is as follows:

$$x_{out}(i)=w_{i,i}x(i)+\sum_{j\in N(i,K)} w_{i,j}x(j) \quad (3)$$

$N(i,K)$ represents the K-hop neighborhoods of the node i. $w_{i,i}$ is the weight combination used.

Spectral graph convolutional

GCN is divided into two models in the literature as spectral-based and spatial-based. Spectral methods can be thought

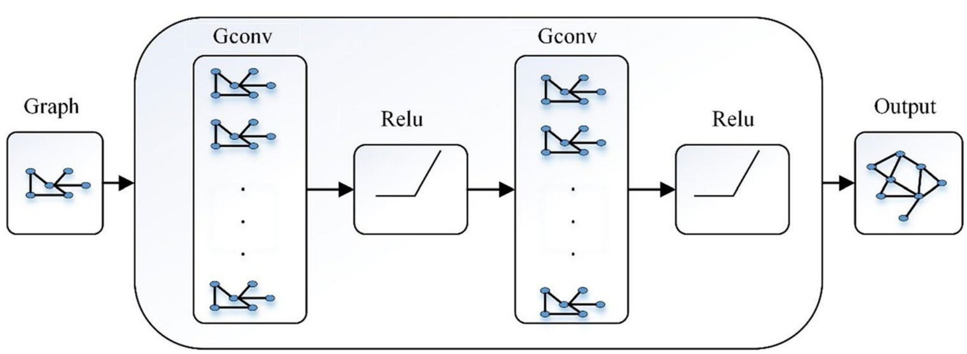


Fig. 6 GCN architecture with a multi-layer structure

of as frequency filtering methods. The first spectral GCN model was influenced by the classic CNN structure (Bruna et al. 2013; Zhang et al. 2019):

$$X^{p+1}(:,j) = \sigma\left(\sum_{i=1}^{d_p} V \begin{bmatrix} (\theta^p_{i,j})(1) & \cdots & 0 \\ \vdots & \ddots & \ddots \\ 0 & \cdots & (\theta^p_{i,j})(1) \end{bmatrix} V^T X^P(:,i)\right), \forall_j = 1,\ldots,d_{p+1}, \tag{4}$$

X^p is a vector and it is an input feature map of the pth layer. Its dimension is $n \times d_p$. $X^p(:,i)(X^{p+1}(:,j))$ is ith and jth dimension of input output feature. $\theta^p_{i,j}$ specifies a vector of learnable parameters of the filter in layer p. Each column of V is the eigenvector value of L. σ(.) denotes the activation function.

Spatial graph convolutional

Spatial convolutional tries to perform convolutional processing directly in the vertex domain. During this process, a node collects information from its neighboring nodes in a recursive manner so that the state of the node is updated. The spatial graph model includes models such as the CNN-based model, propagation-based model, and general framework models The CNN-based model has successful applications in areas such as image classification and object recognition. This model is successful in grid data structures. In the propagation-based structure, a model emerges that collects and disseminates the information of neighboring nodes in vertex domains. In this model, the graph convolutional for the node u in the pth layer has the following representation (Zhang et al. 2019):

$$X^P_{N(u)} = X^P(u,;) + \sum_{v \in N(u)} X^P(v,;) \tag{5}$$

$$X^{P+1}(u,;) = \sigma\left(x^P_{N(U)}\theta^P_{|N(u)|}\right) \tag{6}$$

$\theta^P_{|N(u)|}$ represents a weight matrix for nodes.

In this study, we consider the U and v as two nodes in the graph, and X_U and X_V are feature vectors. The encoder function *Enc* (u) and *Enc* (v), which convert feature vectors to Z_U and Z_V, are defined. Here it is necessary to determine the similarity between the Z_U and $\mathbf{Z}_V$ feature vectors using a metric. GCN applies the neural network for each node vector and obtains a learned node vector. Learning is done per edge by performing a similar operation for the edges. GCN uses pooling to collect information from the edges and forward it to the nodes for prediction. Each item to be pooled is embedded and combined into a matrix. The resulting matrix is gathered using the encoder function. The encoder function handles local network neighbors, aggregate information, and multilayer computation. A graph is created by determining nodes and neighborhoods. After the graph is created using the neighborhood information, the addition process is performed with the help of functions such as sum, average, and maximum using neural networks. In GCN, convolution is the operation of collecting and processing the information of an element's neighbors to update the value of the graph elements. GCN uses the model of forward propagation. In GCN, the inputs are the feature vectors of the nodes. The neural network performs aggregation by taking the feature vectors and then transferring them to the next layer.

Due to the inability of classical machine learning models to adapt to the graph structure in training, the approximately 95,000 users who are not labeled as hateful or normal, and the relationships between users were not evaluated. It should be noted that hateful users are careful not to use words that clearly indicate hate speech in order not to reveal their identity. In addition, users who retweet a hateful person's tweets may develop different behavioral patterns. Information about such behavior is hidden in the relationships between users. Therefore, the proposed GCN model uses user properties and relationships among all users in the dataset, including the GCN model, unannotated users. In this study, the StellarGraph library was used to create the GCN model.

The proposed model is based on a prediction for a node, the properties of the node as well as the properties of its neighbours. That is, it stands out that a hateful user is to connect with other hateful users more possible. While the GCN model trains a node during the training phase, it receives information from its neighbors and does this with a graph convolutional neural network layer. This graph layer embeds by sampling and collecting features from local neighbourhoods of nodes. During training, GCN generates new expressions that will generate hidden representations for nodes that are not present in the network. Figure 7 shows the proposed model architecture.

As seen in Fig. 7, the proposed model consists of a three-stage GCN. The output of the last GCN layer becomes an input to a fully connected layer with ReLu functionality. The output layer has a softmax function to determine class labels based on probabilities. Categorical cross-entropy was used as the loss function. Adam optimizer was used as the optimizer in all layers. GridSearchCV is used to

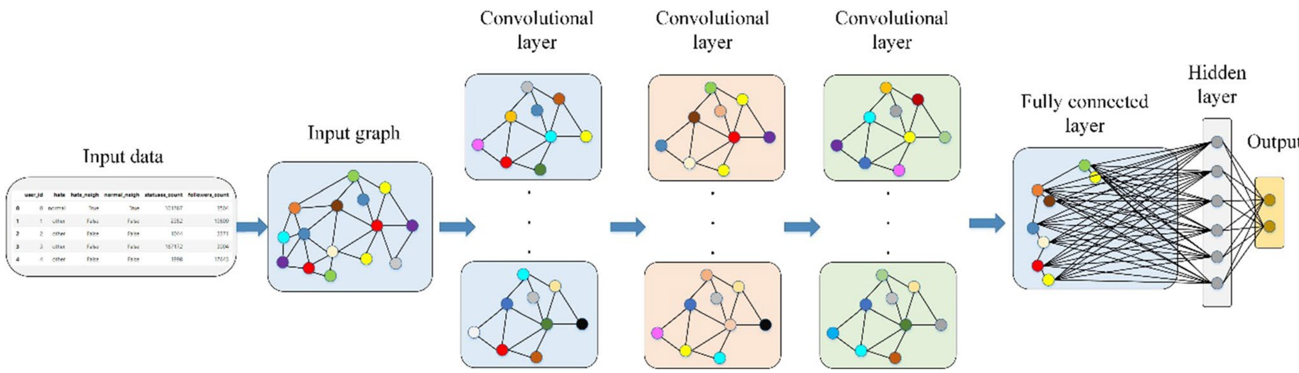


Fig. 7 The proposed model architecture

Table 2 Confusion matrix

		Actual values	
		Positive (1)	Negative (0)
Predicted values	Positive (1)	TP	FP
	Negative (0)	FN	TN

determine pool size, learning rate, layer size, and the number of the epoch. This way, epoch 20, layer sizes [32, 16], and learning rate were selected as 0.005. GCN layers consist of 1024 nodes. fivefold cross-validation was used for model testing.

In the implementation phase of the GCNs model, a StellarGraph object was created to prepare the data for processing using GCNs. Using the Tensorflow Keras library, the Softmax layer is defined to generate the prediction outputs. Adam was used as the optimizer and the binary cross entropy was used for the loss. In order to obtain more successful classification results, hyperparameters such as epoch and learning rate that the model will use were optimized.

Performance evaluation metrics

Classification algorithms aim to classify binary or multiclass categorical values. Basically, accuracy, precision, recall, and F1-score metrics are used to evaluate the errors that occur and to reveal the accuracy rates effectively. Utilizing the values from the confusion matrix, these evaluation metrics are calculated. The outcomes of classification models are interpreted using the confusion matrix to cross-evaluate the errors in the relationship between actual and predicted values. The confusion matrix is shown in Table 2.

Table 1 shows the confusion matrix of the output of a model set up for binary classification. Instances of true positives (TP) are those in which the true value and the predicted value are both 1. True Negatives (TN) are instances where both true and the predicted value is 0. In cases where the true value is 0 but the predicted value is 1, these instances are known as false positives (FP). When the true value is 1 but the predicted value is 0, this is known as a false negative (FN).

According to Eq. 7, accuracy is determined by dividing the number of samples correctly classified by the model by the total number of samples.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{7}$$

How many of the positively predicted values are actually positives is referred to as precision. Equation 8 is used to calculate the precision.

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

Recall is a measure of how many of the outcomes we predict are positive. Equation 9 is used to calculate recall.

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

The F1-score is calculated by taking the harmonic average of precision and recall values. It is a measure of how well the classifier is performing and is often used to compare classifiers. F1-score is calculated as seen in Eq. 10.

$$F - 1\ score = \frac{2 * Precision * Recall}{Precision + Recall} \tag{10}$$

The experimental results

In this study, a deep learning model based on GCN was proposed to detect hateful users. The results obtained from the DT, LR, NB, RF, SVM, MLP, and Bi-LSTM models were used to accurately evaluate the success of the results obtained with the GCN model. Each model's achieved accuracy, precision, recall, and F1-score were compared. To determine the parameters of the used deep learning and machine learning models, parameter analysis studies were conducted in the study utilizing GridSearchCV. Cross-validation was employed in the applied models to solve the overfitting issue and raise the caliber of the models built. Cross-validation allows the performance of the model to be tested before encountering high error rates on an as-yet-unseen dataset. Cross-validation was made by choosing the k value as 5. In the graphs below, the confusion matrix results of the DT, LR, NB, RF, SVM, MLP, and Bi-LSTM algorithms and the newly proposed GCN algorithm are given, respectively, and these graphs show the performance of these methods in detecting hateful users correctly.

As seen in Fig. 8, the number of normal users correctly classified by the DT algorithm is 255 and the number of hateful users correctly classified is 1. In total, it classified 256 users correctly and misclassified 47 users. Also seen in the same figure, the number of normal users correctly classified by the LR algorithm is 258, and the number of hateful users correctly classified is 4. In total, it classified 262 users correctly and misclassified 41 users.

Figure 9 shows the results of the NB and RF algorithms. According to these results, the NB algorithm correctly classified 246 normal users and 1 hateful user. The RF algorithm correctly classified 262 normal users and 8 hateful users. In total, the NB algorithm classified 247 users correctly and misclassified 56 users. The RF algorithm classified 270 users correctly but misclassified 33 users.

Figure 10 demonstrates the confusion matrix results of the SVM and MLP algorithms. In light of these results, it is seen that the SVM algorithm correctly classifies 264 normal

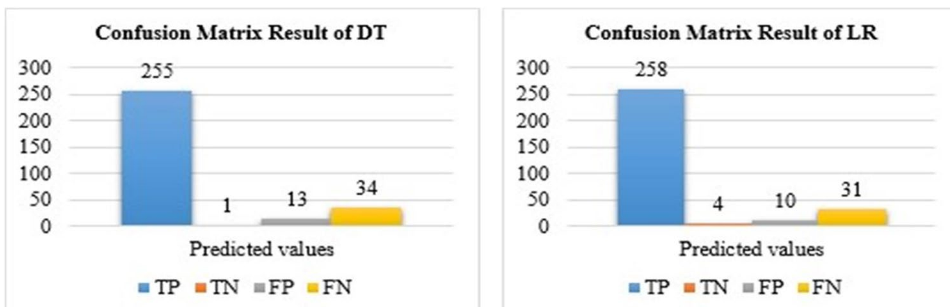


Fig. 8 Confusion matrix results of DT and LR algorithms

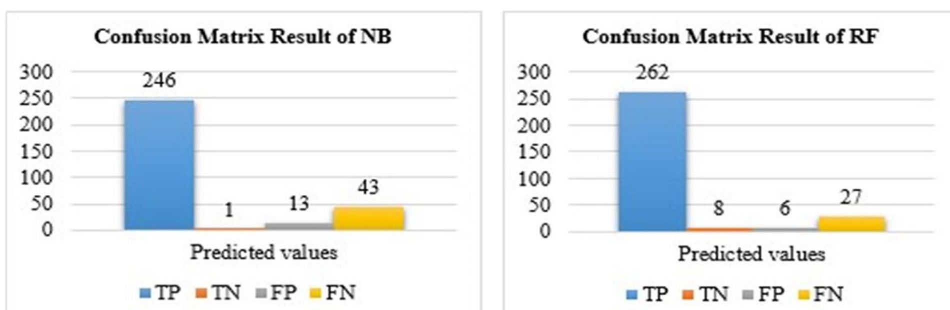


Fig. 9 Confusion matrix results of NB and RF algorithms

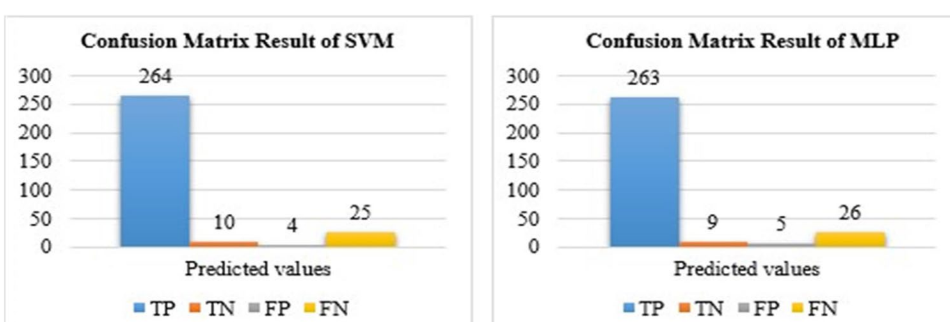


Fig. 10 Confusion matrix results of SVM and MLP algorithms

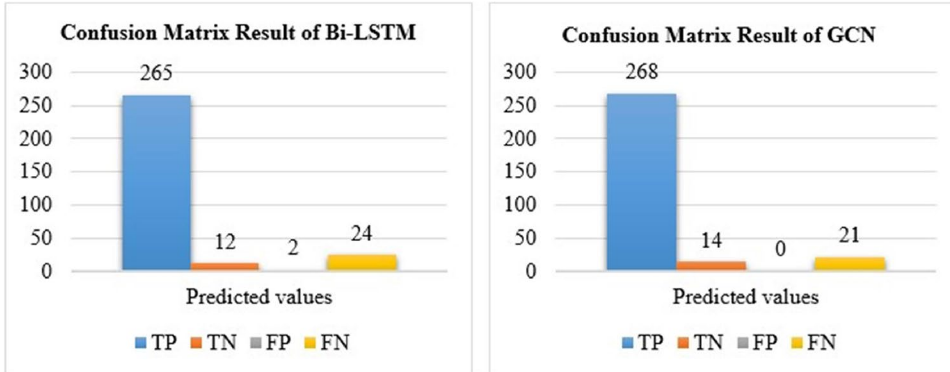


Fig. 11 Confusion matrix results of Bi-LSTM and GCN

Table 3 Comparative experimental results

Model	Accuracy	Precision	Recall	F1-score
DT	0.844	0.906	0.916	0.910
LR	0.864	0.914	0.931	0.922
NB	0.785	0.772	0.981	0.864
RF	0.894	1.000	0.893	0.943
SVM	0.907	1.000	0.905	0.950
MLP	0.897	0.951	0.934	0.942
Bi-LSTM	0.914	0.992	0.916	0.952
GCN	0.930	1.000	0.927	0.962

users and 10 hateful users. It is seen that the MLP algorithm correctly classifies 263 normal users and 9 hateful users. In total, the SVM algorithm classified 274 users correctly and 29 users incorrectly. The MLP algorithm classified 272 users correctly and 31 users incorrectly.

Figure 11 shows the results of the Bi-LSTM and GCN. The Bi-LSTM correctly classified 265 normal and 12 hateful users according to these results. In total, the Bi-LSTM classified 277 users correctly and misclassified 26 users. The GNN method correctly classified 268 normal users and 14 hateful users. In total, it classified 282 users correctly. In addition, the GNN algorithm misclassified 21 users.

Table 3 and Fig. 12 display comparative experimental results for the DT, LR, NB, RF, SVM, MLP, Bi-LSTM, and GCN according to accuracy, precision, recall, and F1-score values.

As demonstrated in Table 3 and Fig. 12, in comparison to the other models, the proposed GCN-based model produced findings that were more favorable. For the proposed model, the accuracy value is 0.930, the precision value is 1.000, the recall value is 0.927, and the F1-score is 0.962.

The proposed GCN model outperformed other models in terms of classification performance for detecting hateful users, as shown in Table 3 and Fig. 12. After the proposed model, Bi-LSTM, SVM, MLP, RF, LR, DT, and NB are the most successful models, respectively.

The proposed model's training and validation accuracy graph was depicted in Fig. 13a, and the training and validation loss graph of the proposed model was depicted in Fig. 13b.

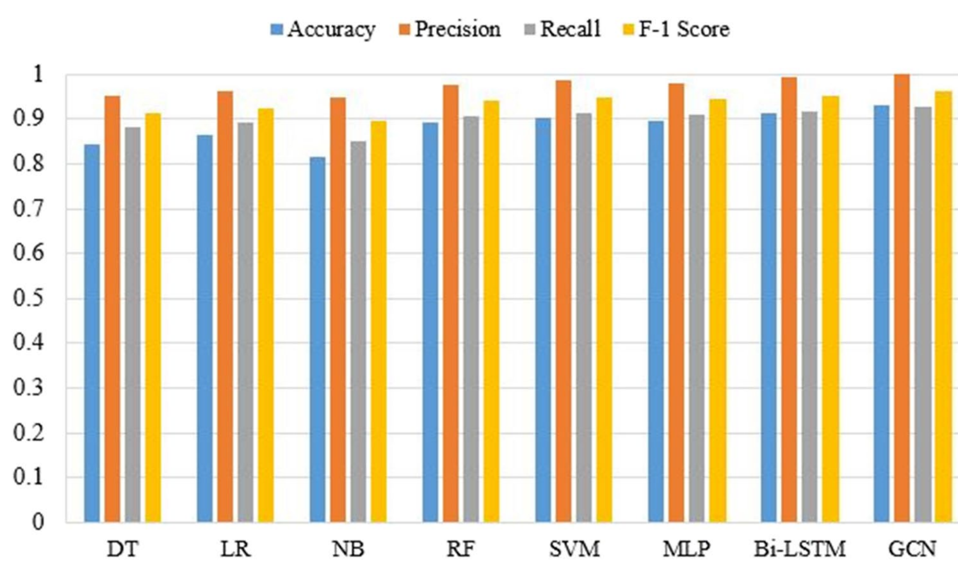


Fig. 12 Comparative experimental results

The ROC curve of the proposed model was demonstrated in Fig. 14. The ROC curve is a metric showing the effectiveness of the developed models using True Positive Rate (TPR) and False Positive Rate (FPR) values. ROC Curve is the curve that shows the values that FPR will take in case TPR increases, that is, in case of 1′ convergence. Here, the convergence of TPR to 1 is a desirable situation. FPR is expected to remain low when the TPR converges to 1.

Conclusions

Today, behavior patterns that include hate speech are common on social networking platforms. For instance, hate speeches about politics, women, foreigners, immigrants, sexual identity-based, beliefs and sect-based, disabled people and various diseases are creating the multidimensional structure of HS. Hate speech is widespread against various segments on online platforms such as Twitter, Facebook, and Instagram, and attacks are organized against individuals or groups targeted through posts. It was observed that these attacks increase the risk of violent tendencies after a certain period of time and that the victim groups are exposed to violent attacks. In other words, hate speech, with its repetitive content and potential violence load, reduces the respect and tolerance towards groups that differ in terms of certain characteristics such as religion, language, race, or sexual orientation, and thus enables the group that creates the discourse to create a new identity value. In societies that are constantly exposed to hate speech, certain stereotypes about minorities begin to form in the public memory, which causes groups that are

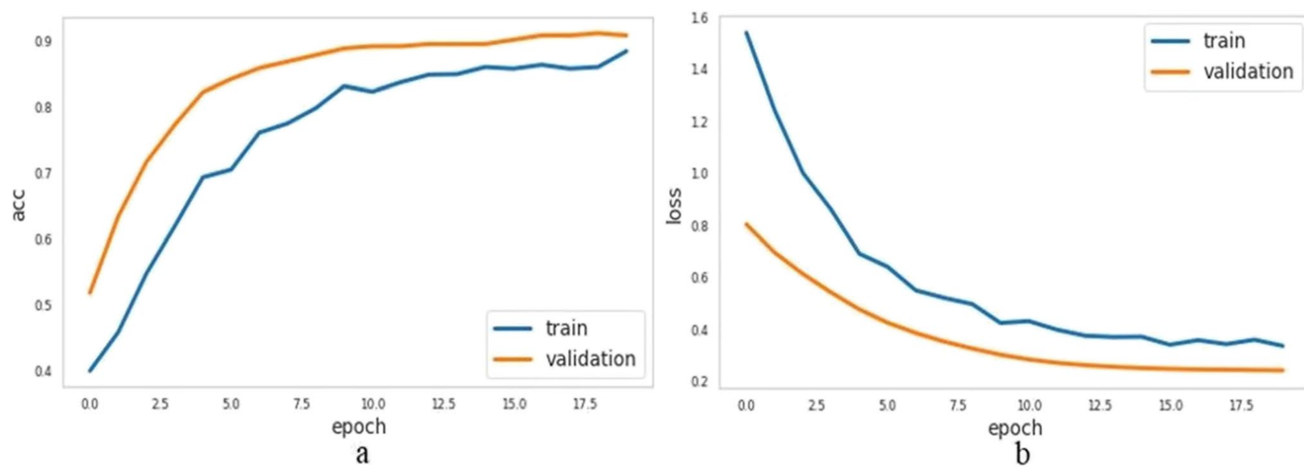
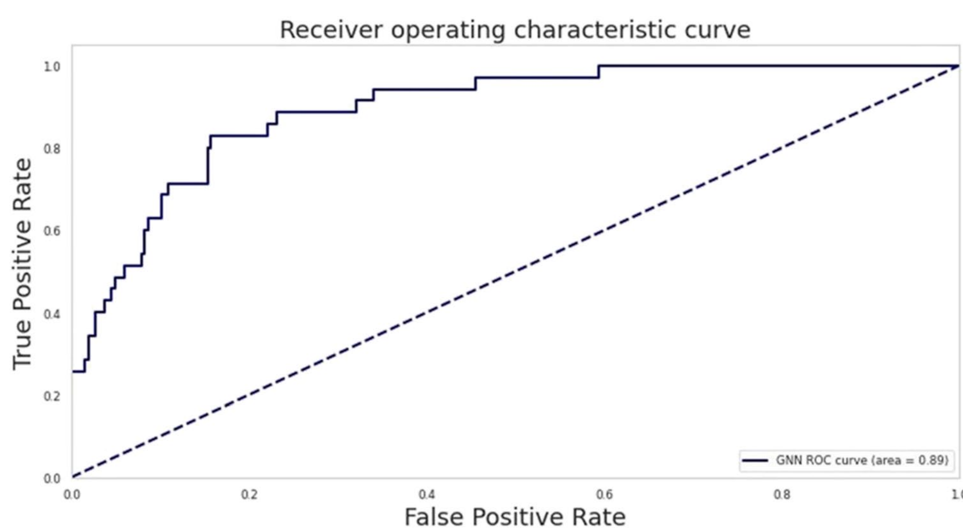


Fig. 13 Training and validation accuracy/loss graphs

Fig. 14 The ROC curve of the proposed model



matched with bad facts to become victims of hate explosions in the future. Since social networking platforms allow interaction between users, the hate speech produced becomes widespread and commonplace in a much more effective way, and in this way, it can be taken for granted and turn into hate crimes over time.

In order to solve the HS problem, intelligent systems with automatic HS detection in social networks are needed. There are many studies in the literature on HS detection, but most of these studies are automatic HS detection systems created using text mining methods (especially NLP-based methods) that analyze users' comments and shares on social networks. In these studies, apart from social network sharing, user features and the relations of these users with each other in a network were not taken into account. However, in our study, besides the analysis of Twitter posts containing hate speech, many features of users such as tweeting frequency, number of followers, number of favorites, and retweeting were used. Hence, hateful Twitter user detection was made with the graph convolutional-based GCN model. In this method, users who spread hate speech were considered a node, and just like in a social network structure, these nodes learn from each other, and HS is detected in the graph structure. This study is one of the few studies using the GCN in this area and compared to popular methods, quite good results were obtained. The GCN method achieved 0.930 accuracy, 1.000 precision, 0.927 recall, and 0.962 F1- score values. As a result of this study, a model proposed that detects users whose HS content is shared on social media platforms in a fast, effective, and automatic way. Thus, users who engage in insults, profanity, humiliation, and harassment can easily be identified. If the user shares a message that contains hate speech, this post may be stopped from publishing, or such behavior may be prevented from happening again. Most importantly, the unforeseen margin of error will be reduced with the automatic HS detection system, and time and cost efficiency will be increased.

Authors contributions All the authors contributed to the study's conception, design, data analysis, and writing of the original manuscript. Anıl Utku carried out data curation, methodology, and software development. Umit Can and Serpil Aslan contributed to the original draft's conceptualization, validation, supervision, rewriting, and editing. All authors read and approved the final manuscript draft.

Data availability Program codes available on request and the datasets generated during and/or analyzed during the current study are available in the Kaggle repository and persistent web links to datasets are below:

https://www.kaggle.com/datasets/manoelribeiro/hateful-users-on-twitter

Declarations

Ethics approval This manuscript has not been published or presented elsewhere in part or in entirety and is not under consideration by another journal. We have read and understood your journal's policies, and we believe that neither the manuscript nor the study violates any of these.

Consent to participate Not applicable.

Consent to publish Not applicable.

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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